HG002 WES Variant Calling and Benchmarking Workflow

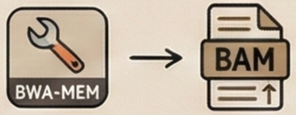
1 Input data

GIAB HG002 WES BAM
Originally aligned to an older reference

2 BAM to FASTQ (initial issue and correction)

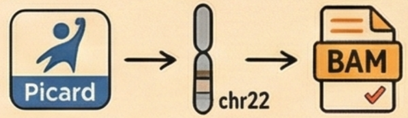
- Initial direct BAM to FASTQ conversion caused paired-read loss due to coordinate sorting
- Fixed by name-sorting BAM before FASTQ extraction

3 Alignment to GRCh38



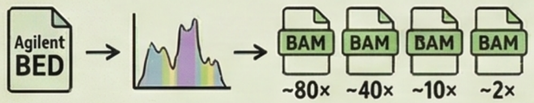
- Alignment using BWA-MEM
- Read groups added
- BAM sorting and indexing

4 Duplicate marking and chr22 subsetting



- Picard MarkDuplicates
- BAM restricted to chromosome 22
- All downstream steps operate on chr22 only

5 Coverage estimation and downsampling



- Depth calculated using Agilent SureSelect V5 exome capture BED
- Baseline depth estimated within capture regions
- Downsampling using Picard DownsampleSam (ConstantMemory) to: ~80×, ~40×, ~10×, ~2×

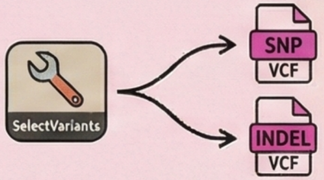
6 Base Quality Score Recalibration (BQSR)

- dbSNP138 used as known sites
- BaseRecalibrator
- ApplyBQSR
- Analyze Covariates

7 Variant calling

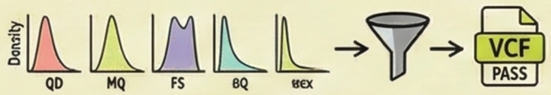
- GATK HaplotypeCaller (GVCF mode, chr22)
- GenotypeGVCFs
- dbSNP rsID annotation retained

8 Variant stratification



- SNP and INDEL separation using GATK SelectVariants

9 Variant quality assessment and filtering



- Density plots for: QD, MQ, MQRankSum, FS, SOR, ReadPosRankSum, DP, InbreedingCoeff
- Stratified by: Coverage, Variant type (SNP, INDEL), Variant novelty (rsID vs novel)
- Coverage-aware filtering using GATK VariantFiltration
- DP thresholds computed as: mean(DP) + 5 × SD(DP)
- PASS-only VCFs generated

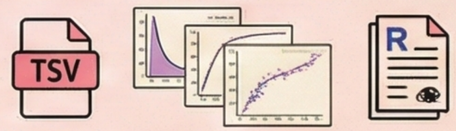
10 Benchmarking preparation

- GIAB HG002 GRCh38 truth VCF
- Confident regions BED
- Intersection with exome capture BED (chr22)
- RTG SDF reference creation
- Dedicated Conda environment (happyenv) for hap.py

11 Benchmarking

- hap.py with vcfeval engine
- Metrics extracted:
  - Precision
  - Recall, F1 score
  - Genotype concordance
  - False positives per megabase

12 Visualization and reporting



- Unified benchmarking TSV
- R-based plots: Precision vs coverage
  - Recall vs coverage, F1 vs coverage
  - Genotype concordance vs coverage,
  - FP per Mb vs coverage
- Final scientific report + technical journey log